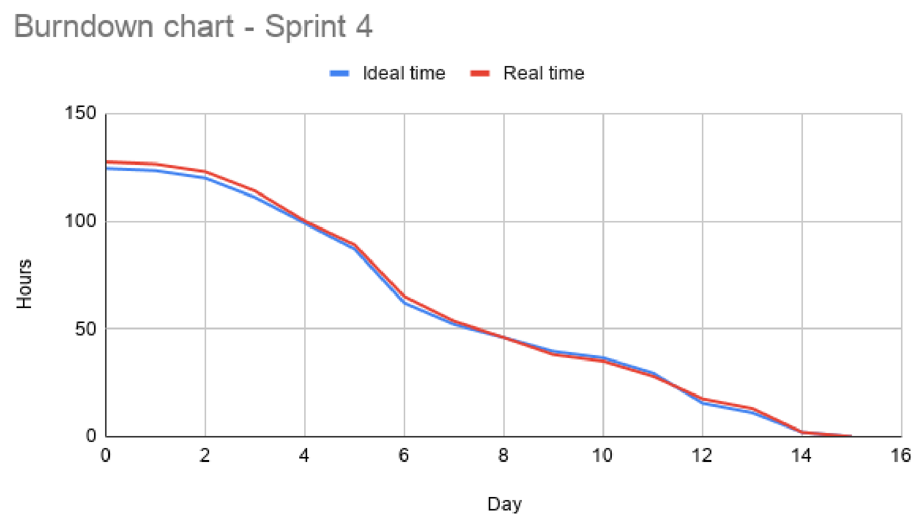
Sprint 4 - Retrospective

Implemented features/completed work + design decisions (narrative)

- Machine learning model development and tests

Burndown chart



The sprint started close to the ideal pace but progress slowed during the middle period, reflecting the complexity of the ML integration and authentication work. The team accelerated toward the end of the sprint and all tasks were completed, though the total hours taken exceeded the original estimate.

Sprint review

Sprint 4 was the final sprint of the project and covered the widest range of tasks across all four sprints. Work was divided into five main areas, covering machine learning development, full completion of the optional features, frontend improvements and refactoring, testing, and the project report.

The machine learning model formed the core technical component of the sprint. Five models were trained and evaluated on the historical bike availability dataset using a 70/30 train-test split, with the random forest regressor selected as the final model based on its performance. The trained model was serialised and integrated into the backend prediction service, with availability predictions displayed through a dedicated panel on the map page. The authentication system and AI chatbot were fully completed during this sprint, with the chatbot connected to the Gemini API. Significant frontend improvements were also made, including replacing the Leaflet map placeholder with Google Maps, route planner UI refinements and responsive design improvements across the application. Unit and acceptance testing was carried out across the backend and frontend, and the project report was written in parallel with development work throughout the sprint.

Sprint retrospective

Sprint 4 was a productive few weeks overall, that focused heavily on machine learning and application tests. The machine learning model was successfully trained with predictions displayed on the map through a dedicated panel on the map page. The chatbot was fully implemented and connected to the Gemini AI API. Authentication was also fully implemented during this sprint. In addition to this, significant refinements were made to the frontend portion of the application. This included polishes to the map page, with the Google map replacing the Leaflet placeholder, route planner UI changes, advanced responsive design and consistency improvements to the website itself. The report was written in parallel with development work, which helped avoid condensing the writing into a short time frame.

The volume of tasks in this sprint was significant, which made prioritisation more challenging than in previous sprints, and some tasks took longer than anticipated, particularly the ML integration and authentication flows. A GitHub login was considered but ultimately not completed due to time constraints.

With the project now in its final refinement stage, the focus is on ensuring all features are stable and well tested before submission. Code documentation and refactoring should be prioritised alongside final report completion. For future projects, we would distribute report writing more evenly across sprints rather than concentrating it at the end, as this added pressure during an already demanding final sprint.